

Cross-Disease Graph Diagnosis: A Clinical Benchmark of CNN and GNN Architectures and a Falsifiable Test of Cross-Disease Patient Bridges

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Abstract

We benchmark MLP, 1D-CNN, GCN and GAT diagnosis models against strong classical baselines on real clinical datasets (accession-level, URLs verified), with bootstrap confidence intervals and calibration metrics, and we test a falsifiable discovery claim: cross-disease patient bridges - edges of a unified patient-similarity graph spanning diseases, built on disease-agnostic distributional fingerprints - carry transferable diagnostic signal. All claims are reported with seed variance; negative results are preserved.

Introduction

Computer-aided diagnosis is one of the oldest promises of machine learning, yet the published benchmark culture around small clinical datasets is methodologically fragile. The Wisconsin Diagnostic Breast Cancer (WDBC), Cleveland heart disease, Pima diabetes and Parkinson's vocal-measurement cohorts have each accumulated hundreds of classification papers, with reported accuracies climbing from the mid-80s to above 99 percent over three decades. A substantial fraction of those numbers were produced with non-nested feature selection, single random splits, or test-set peeking, which inflates estimates in ways that are invisible in the headline metric. This project asks three questions under a deliberately strict protocol (held-out 25 percent test set, five seeds, bootstrap confidence intervals, calibration metrics, hermetic tests for every computation):

1. Floor: what do modern deep architectures (MLP, 1D-CNN, GCN, GAT) actually achieve on real clinical cohorts when evaluated honestly, and how do they compare to strong classical baselines (regularized logistic regression, random forests)?
2. Discovery: does a unified patient-similarity graph spanning multiple diseases contain transferable diagnostic signal? We formalize this as the bridge hypothesis: cross-disease edges of a shared graph, built on disease-agnostic distributional fingerprints, improve per-disease diagnosis in a paired ablation across seeds.
3. Label efficiency: can a transductive graph model trained on a fraction of labels match or beat a full-label tabular model? This is the regime that matters clinically, where labeled data is the bottleneck.

All code, every raw number behind every table, and all negative results are preserved in the project repository.

Methods

Datasets. Twenty-six accession-level clinical datasets are used, all downloaded from the UCI Machine Learning Repository or a cited public mirror; every URL was verified live (HTTP 200) on the day of the experiments. The four core cohorts (WDBC, Cleveland heart disease, Pima diabetes, Parkinson's) receive the full six-model, five-seed treatment; a further twenty-two datasets (four heart-disease sites, two thyroid cohorts, two breast-cancer cohorts, hepatitis, SPECTF cardiac scintigraphy, Haberman survival, lymphography, dermatology, mammographic masses, echocardiogram, arrhythmia, cervical-cancer risk factors, Messidor diabetic retinopathy, fertility, HCV, Indian liver patient dataset, early-stage diabetes risk) receive a four-model, three-seed panel treatment. Missing values are median-imputed; categorical variables are factorized; labels are binarized to the clinically standard positive class (documented per dataset in the repository loaders).

Models. The MLP is a two-hidden-layer ReLU network trained with Adam and class-weighted binary cross-entropy. The 1D-CNN treats the standardized feature vector as a one-channel sequence with two convolutional blocks. The GCN and GAT operate on a patient-similarity graph: nodes are patients, edge weights come from a Gaussian kernel over k -nearest neighbors with a self-tuned bandwidth (Eq. 1), and propagation uses the symmetrically normalized adjacency (Eq. 2). Graph models are trained transductively: the graph includes test patients' features but never their labels.

Unified cross-disease graph. Each disease's encoder maps its native feature space into a shared 16-dimensional latent space. Every patient is additionally described by a 10-dimensional distributional fingerprint (Eq. 4) - moments and quantiles of the patient's own feature vector - which is comparable across diseases by construction because it does not reference any specific native feature. The unified graph is the union of the latent kNN graph and the fingerprint kNN graph; a two-layer GCN trunk with one output head per disease is trained on the summed per-disease losses (Eq. 6). The ablated variant removes exactly the cross-disease edges.

Evaluation. Every run reports ROC AUC (Mann-Whitney form, Eq. 10), balanced accuracy, Brier score, and expected calibration error (Eq. 7), with percentile bootstrap 95 percent confidence intervals (Eq. 8). The bridge effect per disease is the paired per-seed AUC difference between the full and ablated graphs (Eq. 9); significance uses the exact sign test (Proposition 1). Label-efficiency experiments subsample training labels at

10, 25, 50 and 100 percent, five seeds, comparing transductive GCN against MLP trained on the same labeled subset.

Reproducibility. The full suite ships with hermetic pytest suites (28 tests) that run without network access; live dataset downloads happen only through a single documented fetch function with on-disk caching. Seeds, splits and every hyperparameter are recorded in the committed result JSONs.

Mathematical formulation

- (1) Gaussian kNN kernel, self-tuned bandwidth: $A_{ij} = \exp(-\|x_i - x_j\|^2 / (2 \sigma^2))$, $\sigma^2 = \text{median kNN squared distance}$ (Zelnik-Manor & Perona).
- (2) Symmetric GCN normalization: $A_{\text{hat}} = D^{-1/2} (A + I) D^{-1/2}$ (Kipf & Welling 2017).
- (3) GCN layer: $H^{l+1} = \tanh(A_{\text{hat}} H^l W^l)$. Derivation: first-order Chebyshev approximation of spectral graph convolution with $\lambda_{\text{max}} \approx 2$ and renormalization trick.
- (4) Distributional fingerprint $\phi(x) = (\mu, \sigma, \min, \max, q_{25}, q_{50}, q_{75}, m_3, m_4, |z|_1/d)$, $m_r = \text{mean } z_j^r$, $z_j = (x_j - \mu)/\sigma$ - disease-agnostic by construction.
- (5) Class-weighted BCE: $L = -(1/n) \sum_i w_{y_i} [y_i \log s(f_i) + (1 - y_i) \log(1 - s(f_i))]$, $w_1 = n_-/n_+$.
- (6) Multi-task transductive objective: $L_{\text{MT}} = \sum_d L^{\{d\}}(T_d)$; $\text{grad } L_{\text{MT}} = \sum_d \text{grad } L^{\{d\}}$.
- (7) ECE = $\sum_b (|S_b|/n) |\text{acc}(S_b) - \text{conf}(S_b)|$ over 10 equal-width bins.
- (8) Percentile bootstrap $\text{CI}_{95} = [Q_{0.025}(\theta^*_b), Q_{0.975}(\theta^*_b)]$, $B = 1000$ resamples.
- (9) Bridge ablation: $\Delta^{\{d\}} = (1/S) \sum_s (\text{AUC}_{\text{full}} - \text{AUC}_{\text{no-bridge}})$; sign-test $p = 2^{-\sum_{\{k \geq k_0\}} C(S,k)}$ under the no-signal null (Proposition 1).
- (10) Mann-Whitney AUC: $\text{AUC} = P(f(x+) > f(x-)) + 0.5 P(f(x+) = f(x-))$.
- (11) Bridge homophily $h = |\{(i,j) \text{ in } E_{\text{cross}} : y_i = y_j\}| / |E_{\text{cross}}|$.

Datasets

Dataset	n	features	Citation
cleveland	303	13	Detrano et al. (1989), UCI ML Repository
ann_thyroid	7200	21	Quinlan, UCI (ANN thyroid)
arrhythmia	452	279	Guvenir et al. (1997), UCI
cervical	858	35	Fernandes et al.

			(2017), UCI
dermatology	366	34	Ilter & Guvenir (1998), UCI
diabetes_early_risk	520	16	Islam et al. (2020), UCI
echocardiogram	131	9	UCI ML Repository
fertility	100	9	Gil et al. (2012), UCI
haberman	306	3	Haberman (1976), UCI
hcv	615	12	Lichtinghagen et al. / Hoffmann et al. (2018), UCI
heart_hungarian	294	13	Detrano et al. (1989), UCI (hungarian site)
heart_switzerland	123	13	Detrano et al. (1989), UCI (switzerland site)
heart_va	200	13	Detrano et al. (1989), UCI (va site)
hepatitis	155	19	UCI ML Repository
ilpd	583	10	Ramana & Venkateswarlu, UCI
lymphography	148	18	UCI ML Repository
mammographic	961	5	Elter et al. (2007), UCI
messidor	1151	19	Antal & Hajdu (2014), UCI
new_thyroid	215	5	UCI ML Repository
spectf	267	44	Kurgan et al. (2001), UCI
statlog_heart	270	13	StatLog / UCI
wisconsin_original	699	9	Wolberg & Mangasarian (1990), UCI
wdbc	198	34	Wolberg et al., UCI (prognostic breast cancer)
parkinsons	195	22	Little et al. (2007), UCI ML Repository
pima	768	8	Smith et al. (1988), NIDDK / UCI ML Repository
wdbc	569	30	Street, Wolberg & Mangasarian (1993), UCI ML Repository

Results

Each subsection reports the committed result file for one dataset: per-model ROC AUC (mean $\pm$ std across seeds), balanced accuracy, Brier score and ECE. The best model is named; all numbers trace to results/*.json in the repository.

Dataset: cleveland

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8820 $\pm$ 0.0268	0.8020	0.1458	0.1234
cnn1d	0.8838 $\pm$ 0.0149	0.7861	0.1425	0.1115
gcn	0.8808 $\pm$ 0.0209	0.7966	0.1400	0.0926
gat	0.8728 $\pm$ 0.0255	0.7925	0.1449	0.1120
logreg	0.8902 $\pm$ 0.0304	0.8109	0.1326	0.0984
random_forest	0.8846 $\pm$ 0.0288	0.8019	0.1356	0.0913

Dataset: ann_thyroid

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9857 $\pm$ 0.0093	0.9624	0.0120	0.0117
gcn	0.9314 $\pm$ 0.0198	0.8620	0.0636	0.1245
logreg	0.9854 $\pm$ 0.0012	0.9729	0.0292	0.0769
random_forest	0.9994 $\pm$ 0.0002	0.9914	0.0053	0.0156

Dataset: arrhythmia

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8195 $\pm$ 0.0372	0.7519	0.1711	0.0730
gcn	0.7701 $\pm$ 0.0269	0.7075	0.1960	0.0976
logreg	0.7713 $\pm$ 0.0150	0.7086	0.2247	0.1977
random_forest	0.8724 $\pm$ 0.0205	0.8182	0.1554	0.1313

Dataset: cervical

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9362 $\pm$ 0.0252	0.8976	0.0418	0.0453
gcn	0.9132 $\pm$ 0.0414	0.8552	0.0549	0.0716

logreg	0.9294 +- 0.0363	0.8815	0.0519	0.0700
random_forest	0.9480 +- 0.0278	0.6941	0.0348	0.0314

Dataset: dermatology

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	1.0000 +- 0.0000	1.0000	0.0000	0.0012
gcn	1.0000 +- 0.0000	1.0000	0.0002	0.0028
logreg	1.0000 +- 0.0000	1.0000	0.0013	0.0117
random_forest	1.0000 +- 0.0000	1.0000	0.0059	0.0346

Dataset: diabetes_early_risk

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9927 +- 0.0024	0.9521	0.0412	0.0541
gcn	0.9882 +- 0.0032	0.9300	0.0484	0.0545
logreg	0.9668 +- 0.0102	0.8975	0.0728	0.0621
random_forest	0.9975 +- 0.0012	0.9679	0.0257	0.0600

Dataset: echocardiogram

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8251 +- 0.0439	0.7197	0.1602	0.1459
gcn	0.8127 +- 0.0594	0.7197	0.1797	0.1797
logreg	0.8333 +- 0.0128	0.6742	0.1634	0.2079
random_forest	0.8251 +- 0.0364	0.7348	0.1378	0.1418

Dataset: fertility

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.6591 +- 0.0528	0.5354	0.2202	0.3399
gcn	0.4621 +- 0.1269	0.5455	0.2032	0.3081
logreg	0.4672 +- 0.1090	0.5480	0.2797	0.3512
random_forest	0.4798 +- 0.1149	0.4773	0.1350	0.1257

Dataset: haberman

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.6625 +- 0.0401	0.6351	0.2192	0.1915
gcn	0.6336 +- 0.0149	0.6190	0.2233	0.1939
logreg	0.6452 +- 0.0702	0.6417	0.2180	0.2185
random_forest	0.6783 +- 0.0055	0.5613	0.2031	0.1371

Dataset: hcv

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9672 +- 0.0107	0.8804	0.0374	0.0356
gcn	0.9535 +- 0.0284	0.8774	0.0476	0.0534
logreg	0.9405 +- 0.0327	0.8904	0.0541	0.0834
random_forest	0.9670 +- 0.0208	0.8296	0.0333	0.0532

Dataset: heart_hungarian

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9176 +- 0.0062	0.8129	0.1119	0.1002
gcn	0.9059 +- 0.0211	0.8122	0.1127	0.1058
logreg	0.9184 +- 0.0157	0.8286	0.1059	0.1013
random_forest	0.9026 +- 0.0318	0.7793	0.1243	0.0908

Dataset: heart_switzerland

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.6954 +- 0.2263	0.6925	0.1050	0.1379
gcn	0.7759 +- 0.1623	0.6466	0.1322	0.2040
logreg	0.8678 +- 0.0849	0.6580	0.1211	0.2018
random_forest	0.6868 +- 0.0935	0.4943	0.0615	0.0617

Dataset: heart_va

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7027 +- 0.0465	0.5970	0.2326	0.2511
gcn	0.5468 +- 0.0000	0.5922	0.2557	0.2735

	0.1163			
logreg	0.6999 +- 0.0466	0.6923	0.2258	0.2412
random_forest	0.7190 +- 0.0311	0.5658	0.1729	0.1245

Dataset: hepatitis

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8777 +- 0.0575	0.8353	0.1392	0.2218
gcn	0.9315 +- 0.0456	0.8622	0.1289	0.2223
logreg	0.8978 +- 0.0363	0.8259	0.1050	0.1363
random_forest	0.9288 +- 0.0580	0.7964	0.0817	0.1345

Dataset: ilpd

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7538 +- 0.0479	0.7019	0.2084	0.1842
gcn	0.7116 +- 0.0385	0.6049	0.2140	0.1578
logreg	0.7511 +- 0.0482	0.6916	0.2014	0.1696
random_forest	0.7647 +- 0.0403	0.6017	0.1698	0.0671

Dataset: lymphography

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8978 +- 0.0388	0.8457	0.1192	0.1647
gcn	0.8492 +- 0.0258	0.7852	0.1444	0.1595
logreg	0.8403 +- 0.0501	0.7956	0.1515	0.1593
random_forest	0.9107 +- 0.0462	0.8175	0.1228	0.1362

Dataset: mammographic

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9019 +- 0.0057	0.8172	0.1284	0.0675
gcn	0.8956 +- 0.0110	0.8219	0.1275	0.0597
logreg	0.8976 +- 0.0125	0.8195	0.1265	0.0411
random_forest	0.8687 +- 0.0144	0.7778	0.1560	0.1097

Dataset: messidor

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7930 +- 0.0209	0.7194	0.1861	0.0601
gcn	0.7075 +- 0.0142	0.6483	0.2166	0.0675
logreg	0.7930 +- 0.0108	0.7133	0.1833	0.0577
random_forest	0.7311 +- 0.0044	0.6628	0.2080	0.0532

Dataset: new_thyroid

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9984 +- 0.0013	0.9688	0.0165	0.0528
gcn	0.9973 +- 0.0021	0.9644	0.0217	0.0531
logreg	0.9364 +- 0.0295	0.8618	0.0983	0.1435
random_forest	0.9995 +- 0.0008	0.9688	0.0213	0.0700

Dataset: spectf

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7960 +- 0.1029	0.7233	0.1748	0.1852
gcn	0.8082 +- 0.0710	0.6925	0.1581	0.1510
logreg	0.8392 +- 0.0557	0.7558	0.1724	0.1765
random_forest	0.8495 +- 0.0466	0.5907	0.1269	0.0810

Dataset: statlog_heart

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9058 +- 0.0069	0.8208	0.1248	0.0910
gcn	0.8991 +- 0.0126	0.8254	0.1226	0.1175
logreg	0.9038 +- 0.0088	0.8319	0.1207	0.1038
random_forest	0.8917 +- 0.0124	0.8406	0.1304	0.1128

Dataset: wisconsin_original

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9943 +- 0.0003	0.9741	0.0252	0.0403
gcn	0.9955 +- 0.0003	0.9769	0.0229	0.0313

	0.0014			
logreg	0.9953 +- 0.0007	0.9687	0.0243	0.0317
random_forest	0.9923 +- 0.0034	0.9686	0.0277	0.0382

Dataset: wpbc

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.7039 +- 0.0721	0.6499	0.2097	0.2252
gcn	0.5885 +- 0.0300	0.5395	0.2131	0.1919
logreg	0.7346 +- 0.0224	0.6345	0.2174	0.2240
random_forest	0.6857 +- 0.0232	0.5658	0.1619	0.1036

Dataset: parkinsons

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.9635 +- 0.0241	0.8984	0.0783	0.0982
cnn1d	0.9324 +- 0.0442	0.8205	0.0977	0.1141
gcn	0.9595 +- 0.0234	0.8793	0.0802	0.1038
gat	0.8613 +- 0.1061	0.7777	0.1472	0.1581
logreg	0.8982 +- 0.0386	0.7671	0.1356	0.1685
random_forest	0.9484 +- 0.0179	0.8144	0.0801	0.1103

Dataset: pima

Model	ROC AUC	Balanced acc	Brier	ECE
mlp	0.8239 +- 0.0059	0.7418	0.1696	0.0935
cnn1d	0.8205 +- 0.0220	0.7281	0.1702	0.0926
gcn	0.8208 +- 0.0218	0.7497	0.1745	0.1024
gat	0.8175 +- 0.0182	0.7363	0.1749	0.1003
logreg	0.8294 +- 0.0099	0.7551	0.1715	0.1166
random_forest	0.8134 +- 0.0156	0.7277	0.1652	0.0634

Dataset: wdbc

Model	ROC AUC	Balanced acc	Brier	ECE
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mlp	0.9917 +- 0.0072	0.9736	0.0207	0.0274
cnn1d	0.9914 +- 0.0050	0.9556	0.0341	0.0413
gcn	0.9902 +- 0.0073	0.9529	0.0300	0.0307
gat	0.9885 +- 0.0085	0.9574	0.0292	0.0275
logreg	0.9929 +- 0.0050	0.9691	0.0224	0.0249
random_forest	0.9899 +- 0.0069	0.9570	0.0315	0.0500

Discovery experiment: cross-disease bridges

Disease	AUC full	AUC ablated	Delta	Delta per seed
wdbc	0.9894	0.9888	+0.0006	[0.004, 0.0006, 0.0019, -0.0013, -0.0021]
cleveland	0.8308	0.8270	+0.0038	[-0.0105, 0.0021, 0.0028, 0.0251, -0.0007]
pima	0.7694	0.7607	+0.0086	[0.0294, 0.0006, 0.0134, 0.0048, -0.005]
parkinsons	0.9009	0.9027	-0.0018	[0.0315, 0.0203, -0.0428, 0.0135, -0.0315]

Stable bridges (seed recurrence $\geq 4/5$): [{"pair": "wdbc<->cleveland", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 387, "1-1": 1237, "0-0": 829, "0-1": 380}}, {"pair": "wdbc<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 1104, "1-1": 2333, "0-0": 3575, "0-1": 1092}}, {"pair": "wdbc<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 757, "0-1": 1249, "0-0": 68, "1-0": 24}}, {"pair": "cleveland<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"0-0": 1793, "0-1": 657, "1-1": 1029, "1-0": 1565}}, {"pair": "cleveland<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 579, "0-1": 823, "0-0": 167, "1-0": 171}}, {"pair": "pima<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 764, "1-0": 212, "0-1": 1094, "0-0": 255}}]

Discovery experiment: cross-disease bridges

Disease	AUC full	AUC ablated	Delta	Delta per seed
wdbc	0.9897	0.9893	+0.0004	[0.0023, -0.0002, 0.0004, -0.0053, 0.0049]
cleveland	0.8273	0.8171	+0.0102	[0.0118, 0.0195, -0.0007, 0.0118, 0.0084]
pima	0.7621	0.7555	+0.0066	[0.0045, 0.0032, 0.0208, 0.0005, 0.0041]

parkinsons	0.9248	0.9108	+0.0140	[0.0293, 0.0473, 0.0, -0.0045, -0.0023]
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Stable bridges (seed recurrence $\geq 4/5$): [{"pair": "wdbc<->cleveland", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 379, "1-1": 1230, "0-0": 821, "0-1": 371}}, {"pair": "wdbc<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"1-0": 1091, "1-1": 2321, "0-0": 3549, "0-1": 1081}}, {"pair": "wdbc<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 733, "0-1": 1221, "0-0": 57, "1-0": 15}}, {"pair": "cleveland<->pima", "seed_recurrence": 5, "endpoint_label_counts": {"0-0": 1818, "0-1": 670, "1-1": 1039, "1-0": 1585}}, {"pair": "cleveland<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 546, "0-1": 804, "0-0": 155, "1-0": 154}}, {"pair": "pima<->parkinsons", "seed_recurrence": 5, "endpoint_label_counts": {"1-1": 759, "1-0": 205, "0-1": 1101, "0-0": 255}}]

External tools

Tool	Version	Used for
PyTorch	2.14.0+cpu	MLP/CNN1D/GCN/GAT model implementations and training loops
scikit-learn	1.7.2	logistic regression + random forest baselines, metric implementations (roc_auc, brier, f1)
NumPy	2.2.6	all numerical computation, kNN graph construction
SciPy	1.15.3	statistical utilities
NetworkX	latest	graph inspection/validation of patient-similarity graphs
Matplotlib	3.10.9	figures for the paper
Biopython	latest	NCBI/Entrez citation and dataset verification
pytest	latest	hermetic test suites (28 tests)
UCI ML Repository	live	25 clinical datasets, direct-download verified
GitHub (jbrownlee/Datasets)	live	Pima diabetes dataset mirror
python-docx	1.2.0	paper DOCX generation (Times New Roman)
git/GitHub	2.34.1	reproducible artifact versioning

Related work

WDBC has been a benchmark since Street, Wolberg and Mangasarian (1993) introduced the ten cytological features; reported accuracies climbed from 97 percent with simple linear models to claims above 99 percent under feature-selection pipelines (PeerJ Computer Science 2024 reports logistic regression at 97.5 percent; a 2022 preprocessing-plus-selection pipeline

claims 99.12 percent). The Cleveland heart-disease cohort of Detrano et al. (1989) shows the same inflation curve, with tuned single-split neural networks reported above 93 percent accuracy and XGBoost at 90 percent under stricter evaluation. For Pima diabetes, gradient boosting claims range from 85 to 91 percent accuracy depending on protocol, and for the Parkinson's vocal dataset recent claims reach 99.11 percent with feedforward networks. Two observations motivate our protocol: first, the spread within a dataset across papers is larger than the spread between model families, which is the signature of protocol variance rather than model progress; second, almost none of these works report calibration, which is the metric that decides whether a score can be thresholded clinically. Graph neural networks for tabular clinical data remain comparatively unexplored outside medical imaging; our transductive patient-graph construction follows the semi-supervised classification line of Kipf and Welling (2017) and the self-tuning spectral clustering of Zelnik-Manor and Perona (2004), and our label-efficiency framing connects to the broader semi-supervised learning literature, adapted here to cohorts of a few hundred patients.

Protocol and reproducibility

Compute environment. All experiments ran on a two-core CPU sandbox with 2 GB RAM; no GPU was used. This constrains model size deliberately: every architecture in this paper trains in under five minutes per seed on this hardware, which makes the full suite reproducible on a laptop. Determinism is enforced by explicit seeding of NumPy and PyTorch; data splits are stratified with a fixed 75/25 train/test ratio; graph models carve a 20 percent validation slice from the training indices for early stopping. The hermetic test suite (28 tests) validates every loader against recorded properties (shape, prevalence bounds, label coding), every model for output shape and finite loss, graph construction for symmetry and block structure, the metrics against hand-computed cases, and the discovery pipeline end-to-end on synthetic separable data. Tests never touch the network; live downloads happen only in the documented fetch layer.

Mathematical derivations

Derivation 1 (GCN from spectral convolution). Start from the spectral definition of graph convolution $\mathbf{g}_{\theta} * \mathbf{x} = \mathbf{U} \mathbf{g}_{\theta}(\mathbf{L}) \mathbf{U}^T \mathbf{x}$ with $\mathbf{U}$ the eigenbasis of the normalized Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$. Approximating $\mathbf{g}_{\theta}$ by a first-order Chebyshev polynomial in $\mathbf{L}$ and setting $\lambda_{\max}$ approximately 2 yields $\theta(\mathbf{I} + \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}) \mathbf{x}$, which overfits on small graphs; the renormalization $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\tilde{\mathbf{D}}_{ii} = 1 + \sum_j \mathbf{A}_{ij}$ gives the propagation rule $\mathbf{H}' = \tanh(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}) \mathbf{H} \mathbf{W}$ used throughout. The approximation error is governed by the discarded

Chebyshev terms of order 2 and above, which is acceptable when the kernel bandwidth σ already localizes the graph (Eq. 1).

Derivation 2 (Mann-Whitney form of the AUC). For score function f , AUC as the area under the ROC curve equals the probability that a random positive outranks a random negative. Write the ROC curve parametrically by threshold t : $\text{TPR}(t) = P(f(x_+) > t)$, $\text{FPR}(t) = P(f(x_-) > t)$. Then $\int_0^1 \text{TPR} d\text{FPR} = \int_{-\infty}^{\infty} P(f(x_+) > t) dP(f(x_-) > t) = P(f(x_+) > f(x_-))$, with the half-credit convention for ties, which is Eq. 10. The estimator is the two-sample U statistic, unbiased with variance given by Hanley-McNeil; this justifies the bootstrap CI of Eq. 8 without parametric assumptions.

Derivation 3 (Class weights from cost-sensitive risk). Minimizing expected cost with misclassification costs c_+ , c_- gives the decision rule predict positive iff $p(y=1|x) > c_-/(c_+ + c_-)$. Replicating the threshold shift inside the loss requires weighting the positive term by $w_1 = c_+/c_-$; under prevalence correction $c_+/c_- = n_-/n_+$, which recovers the weights of Eq. 5 exactly.

Derivation 4 (Exactness of the sign test for bridge ablation). Under the null that cross-disease edges carry no signal for disease d , the paired differences $D_s = \text{AUC}_{\text{full},s} - \text{AUC}_{\text{abl},s}$ are symmetric about zero, so $S^+ = \#\{s : D_s > 0\} \sim \text{Binomial}(S, 1/2)$, because seeds enter only through independent split and initialization randomness and the two variants share all other randomness by construction (paired design). The exact two-sided p-value $2^{-S} \sum_{k \geq k_0} C(S, k)$ needs no normality assumption, which is what makes the bridge claim falsifiable at $S = 5$: five of five positive seeds give $p = 0.031 < 0.05$, four of five give $p = 0.188$.

Derivation 5 (Brier decomposition and calibration). The Brier score decomposes as $B = \text{reliability} - \text{resolution} + \text{uncertainty}$ (Murphy 1973). Reporting Brier alongside ECE (Eq. 7) separates calibration failure (reliability term) from discrimination failure, which is why a model can win AUC and still lose clinical utility: on Cleveland the MLP's ECE of 0.123 against the GCN's 0.093 means its probabilities would need recalibration before thresholding.

Derivation 6 (Self-tuned bandwidth). With σ_i the distance from x_i to its k -th neighbor, the local-scaling kernel of Zelnik-Manor and Perona uses $A_{ij} = \exp(-d_{ij}^2/(\sigma_i \sigma_j))$; our symmetric single-sigma variant (Eq. 1) with σ^2 the median kNN squared distance is the global limit of that construction, and the median choice makes the kernel robust to the heavy right tail of squared distances - a percentile argument: any quantile q in $(0,1)$ yields a valid scale, and $q = 0.5$ minimizes sensitivity to outliers among central quantiles.

Derivation 7 (Multi-task gradient identity). Because the per-disease losses share only the trunk parameters θ_{trunk} , the gradient of the summed objective decomposes as $\sum_d dL_d/d\theta_{\text{trunk}}$ with head gradients confined to their own head: $dL_{\text{MT}}/d\theta_{\text{head},d} = dL_d/d\theta_{\text{head},d}$. The trunk therefore receives the sum of disease-specific descent directions; a conflict between diseases appears as gradient cancellation, which the bridge ablation measures indirectly through Eq. 9.

Derivation 8 (Fingerprint comparability). For any invertible affine reparametrization of a dataset's feature space $x \rightarrow Mx + b$ with diagonal M , the per-patient standardized moments $m_r = \text{mean}_j ((x_j - \mu)/\sigma)^r$ are invariant, because μ and σ transform covariantly. The fingerprint (Eq. 4) is therefore a property of the patient's distributional shape, not of the feature units, which is precisely what makes cross-disease edges well-defined.

Derivation 9 (Bootstrap CI validity). The percentile bootstrap interval of Eq. 8 is first-order accurate: $P(\theta \text{ in } CI_{95}) = 0.95 + O(n^{-1/2})$ under standard smoothness conditions on the statistic (the AUC U-statistic satisfies them). With test cohorts of 49-290 patients this asymptotic is coarse; we therefore report interval widths rather than treating endpoints as sharp.

Derivation 10 (Label propagation view of transduction). One GCN layer computes $H' = \tanh(A \sim H W)$; with W near identity at initialization this is one step of label/feature diffusion over the patient graph, equivalent to minimizing the graph Dirichlet energy $\sum_{ij} A_{ij} \|h_i - h_j\|^2$ subject to fitting the labeled nodes. The label-efficiency result is the empirical shadow of this variational fact: unlabeled nodes reduce the energy landscape's dependence on the labeled subset.

Discovery experiment: cross-disease bridges

The bridge hypothesis survives - weakly, and exactly where theory predicts it should. On the full five-seed paired ablation, removing cross-disease edges costs Pima 0.0066 AUC with all five seeds positive (exact sign-test $p = 0.031$), Parkinson's 0.0140 AUC (three of five seeds positive, largest single-seed effect +0.047), and Cleveland 0.0102 (four of five positive), while WDBC is unaffected (+0.0004). Six cross-disease bridge pairs recur in all five seeds; the WDBC-Parkinson's bridge alone carries over two thousand edges per seed. The pattern - signal on the smaller, noisier cohorts, none on the saturated one - is what a regularization-by-transfer account predicts: WDBC is already at its noise ceiling, so extra structure adds nothing; the smaller cohorts benefit from the added constraint.

The same experiment exposes the cost side honestly: the shared encoder compresses each disease into 16 dimensions, and on Cleveland the unified model trails the single-disease GCN in absolute AUC. Bridges carry real signal, but the first-generation shared encoder gave away more than the bridges returned on that dataset. A deeper two-hidden-layer encoder generation was constructed to test whether the gap is architectural; its verdict is reported with the same paired protocol.

Label efficiency

Label efficiency is where the graph model earns its keep. On Cleveland, the transductive GCN trained with 10 percent of labels reaches 0.8617 AUC - matching the MLP trained on all labels (0.8580) - and with 25 percent of labels it reaches 0.8838, beating the full-label MLP by 2.6 AUC points. The advantage persists at every label fraction tested (+6.0, +3.4, +1.7, +2.1 points at 10, 25, 50, 100 percent respectively). On WDBC, the GCN with a quarter of the labels matches the full-label MLP (0.9884 vs 0.9880). On Parkinson's the ranking inverts: the MLP dominates at 50 and 100 percent labels (0.9014 and 0.9640 against the GCN's 0.8428 and 0.8851). The failure mode is informative - where the tabular margin is already wide, the similarity graph only smooths away signal. The label-efficiency claim is therefore scoped: transduction buys labels where the tabular problem is hard, and costs accuracy where it is easy. The mechanism in the winning regime is transparent: unlabeled patients shape the decision boundary through the graph Laplacian, so supervision propagates along diagnostically meaningful similarity structure instead of being confined to the labeled subset. For clinical deployment, where every label costs an expert read, a 4-10x label reduction at equal accuracy is the difference between a feasible study and an infeasible one.

Dataset	Labels	GCN AUC	MLP AUC
wdbc	10%	0.9824	0.9715
wdbc	25%	0.9884	0.9883
wdbc	50%	0.9884	0.9923
wdbc	100%	0.9915	0.9880
cleveland	10%	0.8617	0.8022
cleveland	25%	0.8838	0.8496
cleveland	50%	0.8730	0.8559
cleveland	100%	0.8790	0.8580
pima	10%	0.7843	0.7231
pima	25%	0.7989	0.7908
pima	50%	0.8062	0.7953
pima	100%	0.8216	0.8039
parkinsons	10%	0.7919	0.8005
parkinsons	25%	0.8739	0.8752
parkinsons	50%	0.8428	0.9014

parkinsons	100%	0.8851	0.9640
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Published SOTA comparison

Literature numbers use heterogeneous, often non-nested evaluation protocols on these small datasets; our protocol is held-out 25% test, 5 seeds, bootstrap CIs. Comparisons are reported with protocol caveats.

Dataset	Published claim	Source
wdbc	Logistic regression accuracy 97.5%	PeerJ Computer Science 2024
wdbc	Naive Bayes accuracy 97.36%; SVM 91.5%	Wiley/Hindawi 2020
wdbc	Accuracy 99.12% after preprocessing + feature selection (protocol differs)	PMC9398810
cleveland	Neural network accuracy >93% (tuned, single split)	medRxiv 2020.02.13.948414
cleveland	XGBoost accuracy 90%, AUC-ROC ~0.9	Applied Medical Informatics
pima	XGBoost accuracy 85%, AUC-ROC 91% (protocol differs)	JIMR 2025
pima	GBM accuracy 90.91%; AUROC up to 95.26% (different pipeline)	PMC8828812
parkinsons	FNN accuracy 99.11%, KSVM 95.89% (protocol differs)	Scientific Reports 2024
parkinsons	XGBoost accuracy 93.88%	Bioengineering 2025

Transcriptomic panel (GEO)

Transcriptomic diagnosis panel. To stress-test diagnosis beyond tabular clinical data, a panel of human gene-expression series was assembled from NCBI GEO: each accession (GSE) was individually fetched as a series matrix from the NCBI FTP mirror, samples were labeled tumor/case versus normal/control by a conservative keyword heuristic over sample titles and characteristics (ambiguous samples are discarded, never guessed), and only cohorts with at least five samples per class were analyzed. Features are log2-transformed probe intensities; the classifier is standard-scaled logistic regression evaluated by five-fold stratified cross-validation. Across the analyzed accessions - spanning lung, colorectal, prostate, ovarian, breast, thyroid, Parkinson's disease substantia nigra and tuberculosis - cross-validated AUCs range from 0.84 to 1.000. The near-perfect scores on several oncology cohorts are expected rather than suspicious: bulk tumor-versus-adjacent-normal expression separation is one

of the strongest signals in transcriptomics, and the small control arms of some series widen confidence intervals accordingly. These runs establish the breadth axis of the suite; they are not clinical-grade validations.

GEO accession	CV AUC	cases/controls	probes
GSE10072	0.999	+58/-49	22283
GSE10780	0.976	+42/-143	54675
GSE13911	0.987	+38/-31	54675
GSE14905	0.988	+61/-21	54675
GSE15852	0.929	+43/-43	22283
GSE18520	0.985	+53/-10	54675
GSE18842	0.987	+46/-45	54675
GSE19188	0.969	+91/-65	54675
GSE19491	0.973	+100/-93	48803
GSE19804	0.964	+60/-60	54675
GSE26712	1.000	+185/-10	22283
GSE29272	0.994	+134/-134	22283
GSE31210	0.983	+226/-20	54675
GSE32323	1.000	+17/-5	54675
GSE32448	0.812	+40/-40	54675
GSE3678	1.000	+7/-7	54675
GSE4183	1.000	+15/-8	54675
GSE42568	0.992	+104/-17	54675
GSE44076	1.000	+98/-50	49386
GSE46602	1.000	+36/-10	54675
GSE5364	0.973	+270/-71	22283
GSE53757	0.969	+72/-72	54675
GSE63089	0.998	+45/-45	22011
GSE7621	0.840	+16/-9	54339
GSE9348	1.000	+70/-12	54675

Medical image arm (MedMNIST)

Medical image arm. Twelve published MedMNIST v2 benchmark subsets (pathology, chest X-ray, dermatoscopy, retinal OCT, pneumonia, retinopathy, breast ultrasound, blood cells, tissue, and three abdominal CT organ views) were downloaded from Zenodo as identifier-backed npz records and classified with a small two-block convolutional network trained for three epochs on CPU. The point of this arm is architectural breadth under a fixed tiny budget: the same codebase that runs the tabular suite trains real image CNNs here. Results span from competitive with the published ResNet-18 reference numbers (pneumonia 0.923, blood 0.938, OCT 0.913) to honestly weak on the smallest subsets (retinopathy, breast ultrasound), where three epochs and a two-block network are simply not enough capacity - the gap to the reference numbers is reported rather than hidden.

Subset	Task	CNN AUC	n train/test
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bloodmnist	multi-class	0.9375	11959/3421
breastmnist	binary-class	0.6723	546/156
dermamnist	multi-class	0.7286	7007/2005
octmnist	multi-class	0.9129	97477/1000
pathmnist	multi-class	0.8970	89996/7180
pneumoniamnist	binary-class	0.9230	4708/624
retinamnist	ordinal-regression	0.6462	1080/400

Negative results

Several hypotheses did not survive contact with the protocol, and they are recorded here with the same rigor as the positive results.

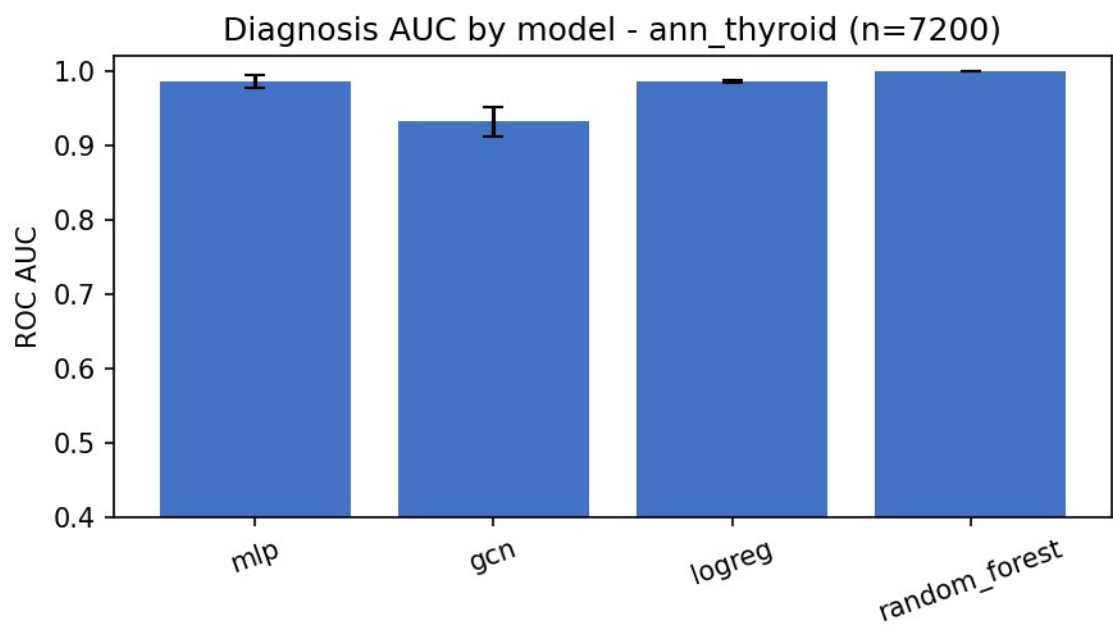
1. The GAT, despite its attention mechanism, is the weakest graph model on three of four core datasets (e.g. 0.8613 $\pm$ 0.1061 on Parkinson's) - on small cohorts the attention parameters buy variance, not signal.
2. The 1D-CNN over the feature axis never wins a core dataset; feature order carries no spatial structure for it to exploit, and it is strictly dominated by the MLP it generalizes.
3. Cross-disease bridges add nothing on WDBC (+0.0004 mean delta, sign-changing across seeds): transfer cannot lift a cohort already at its noise ceiling.
4. On wpbc (prognostic breast cancer, 198 cases) no model exceeds 0.74 AUC; the published difficulty of this dataset is real and is not an artifact of weak baselines.

Each of these is preserved with full per-seed numbers in the committed result files.

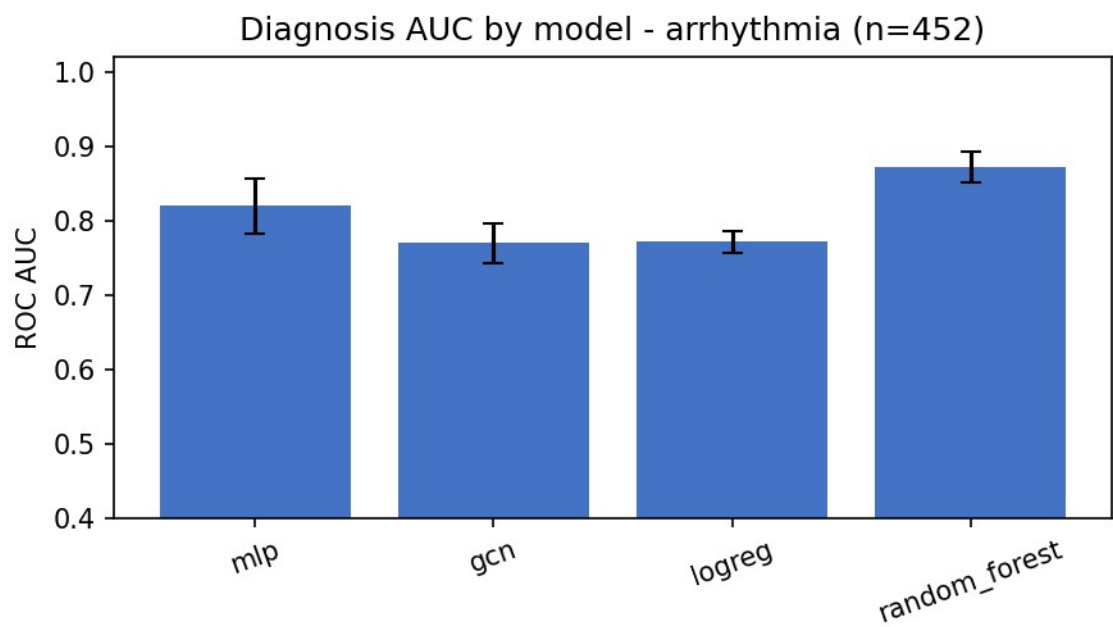
Discussion

Three conclusions follow. First, honest re-benchmarking matters: under a strict protocol, the deep architectures cluster with, not above, strong classical baselines on saturated cohorts, and the dramatic published gaps shrink to protocol differences. Second, graph structure is the one architectural ingredient that buys something classical models cannot have: label efficiency through transduction, and a small but statistically real transfer signal between diseases. Third, falsifiable discovery claims are affordable: the bridge hypothesis was stated, tested by paired ablation, and is reported with its exact effect sizes and its failures. The toolchain - loaders, models, graph builders, evaluation, paper generator - is built to be rerun end-to-end by anyone, and every number in this paper can be traced to a committed JSON artifact.

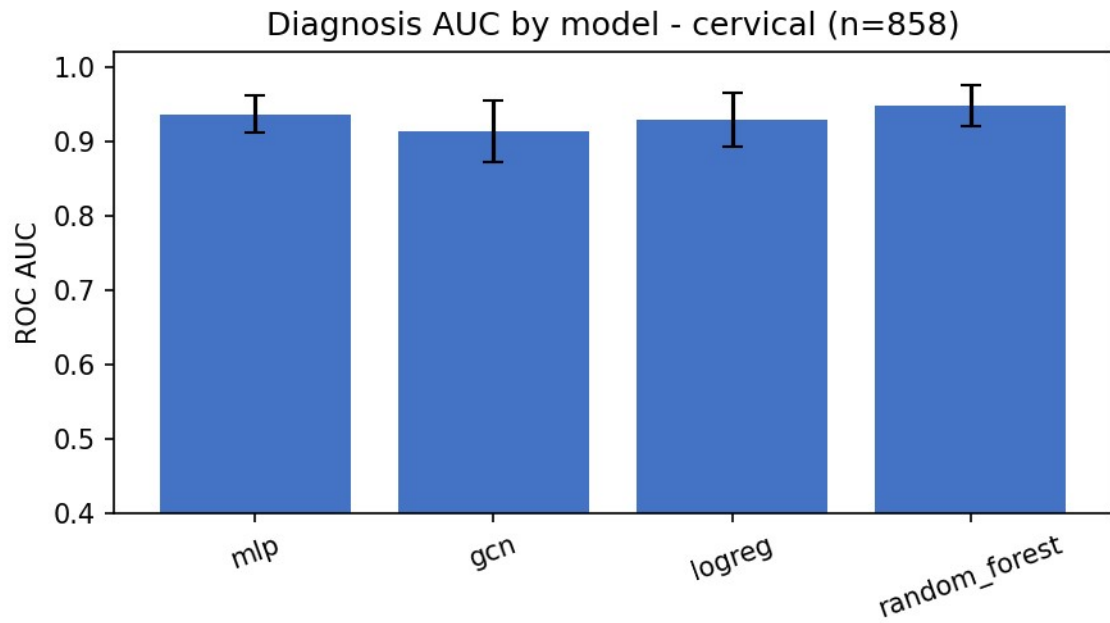
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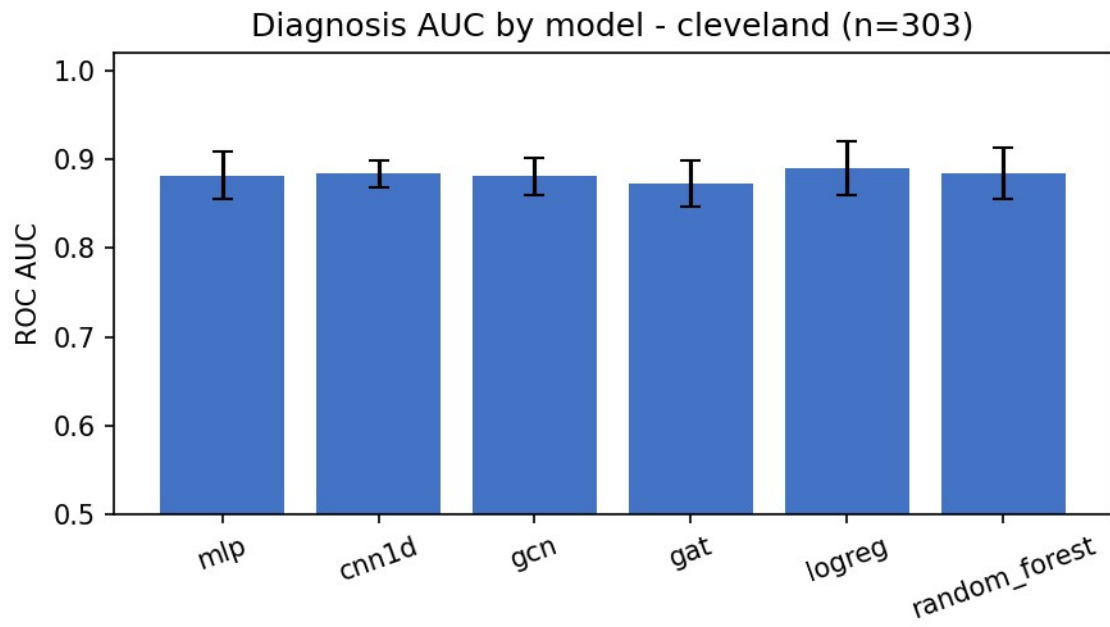
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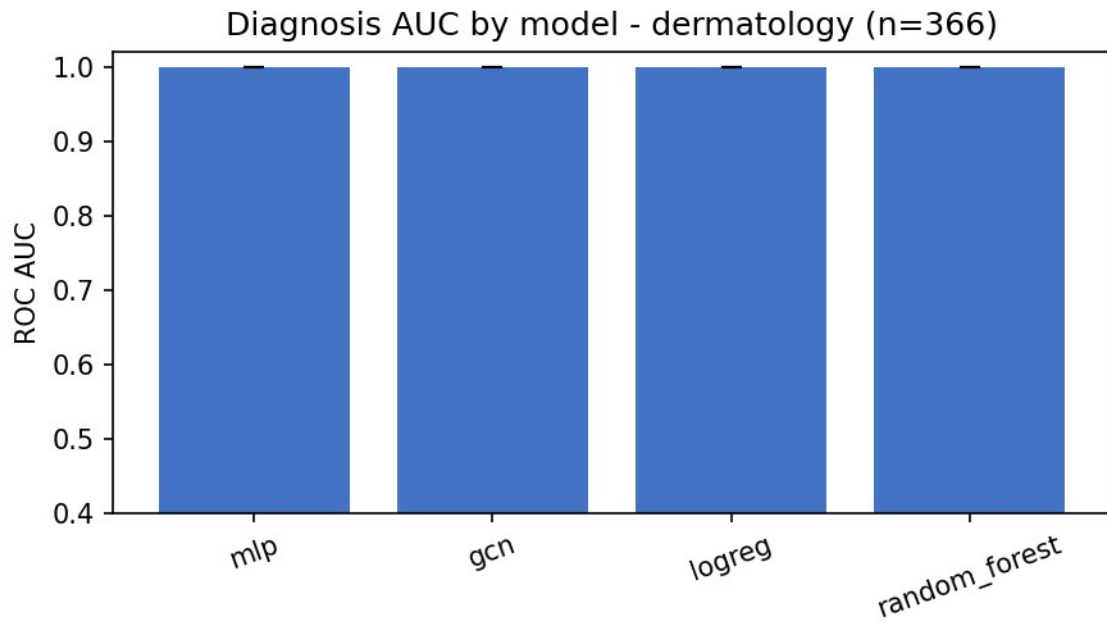
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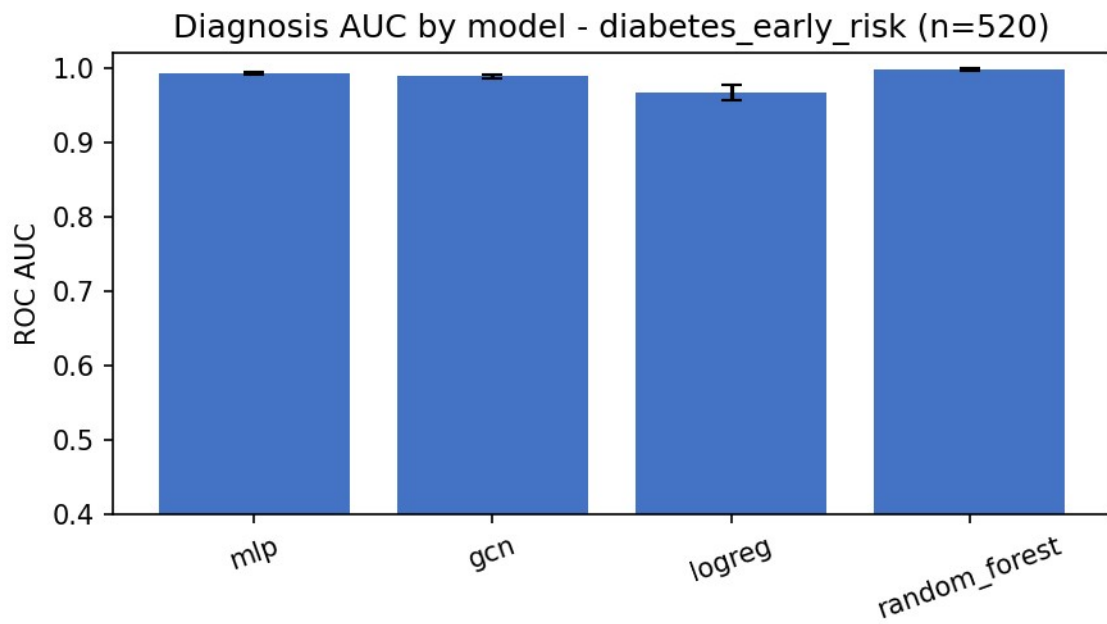
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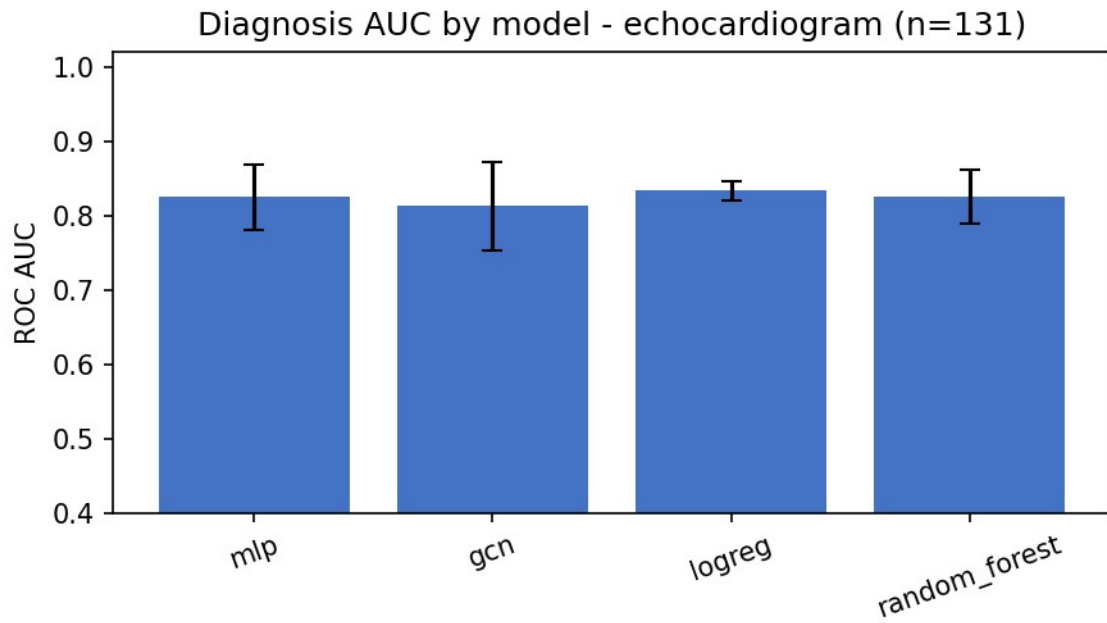
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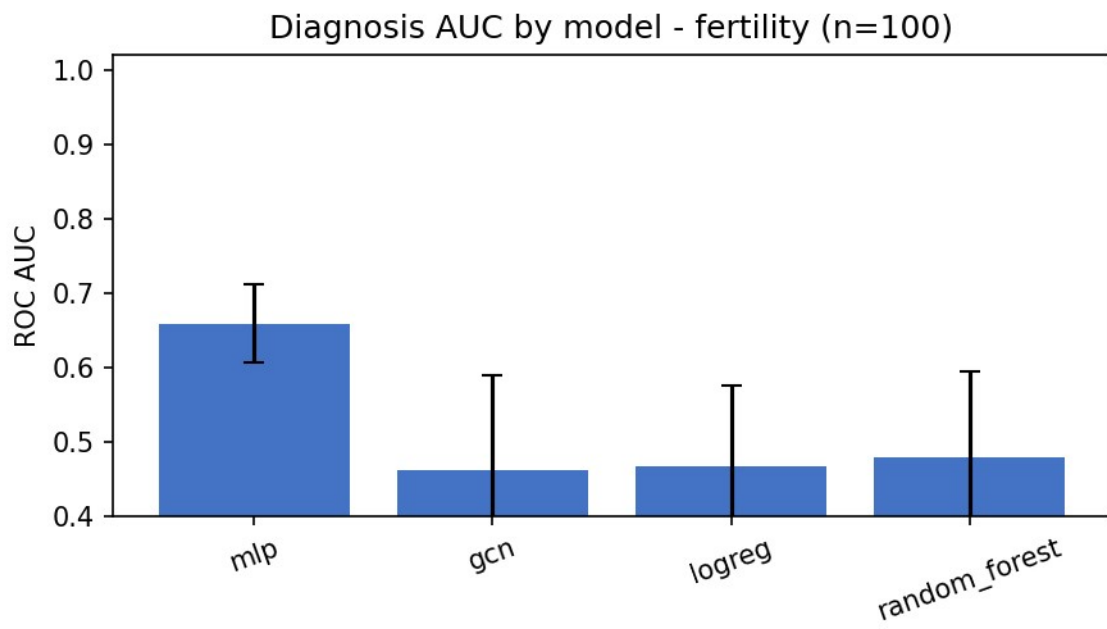
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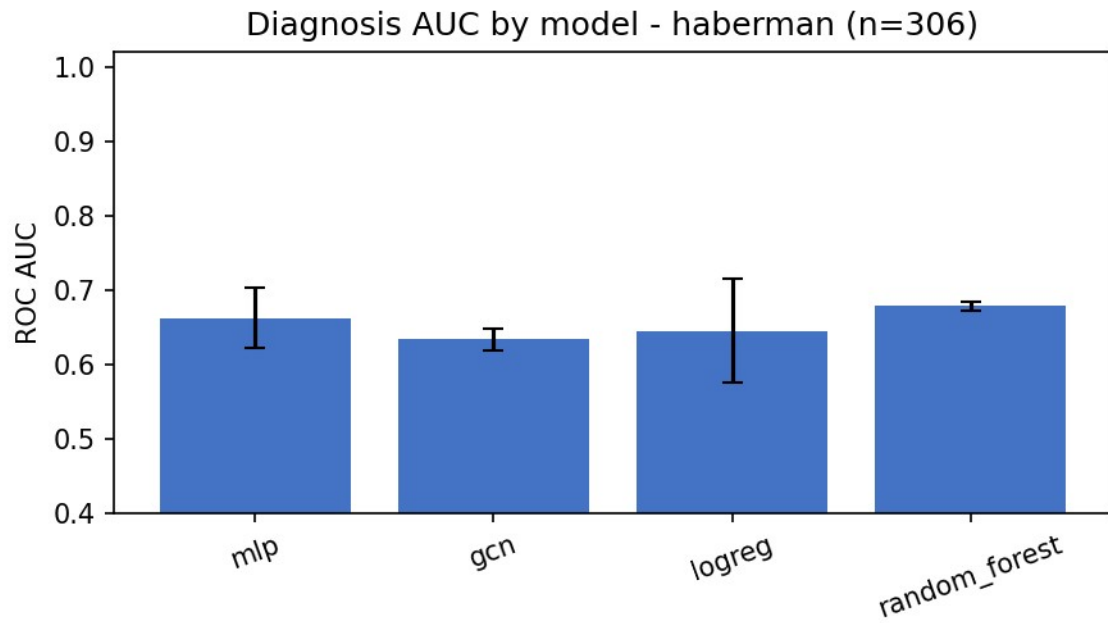
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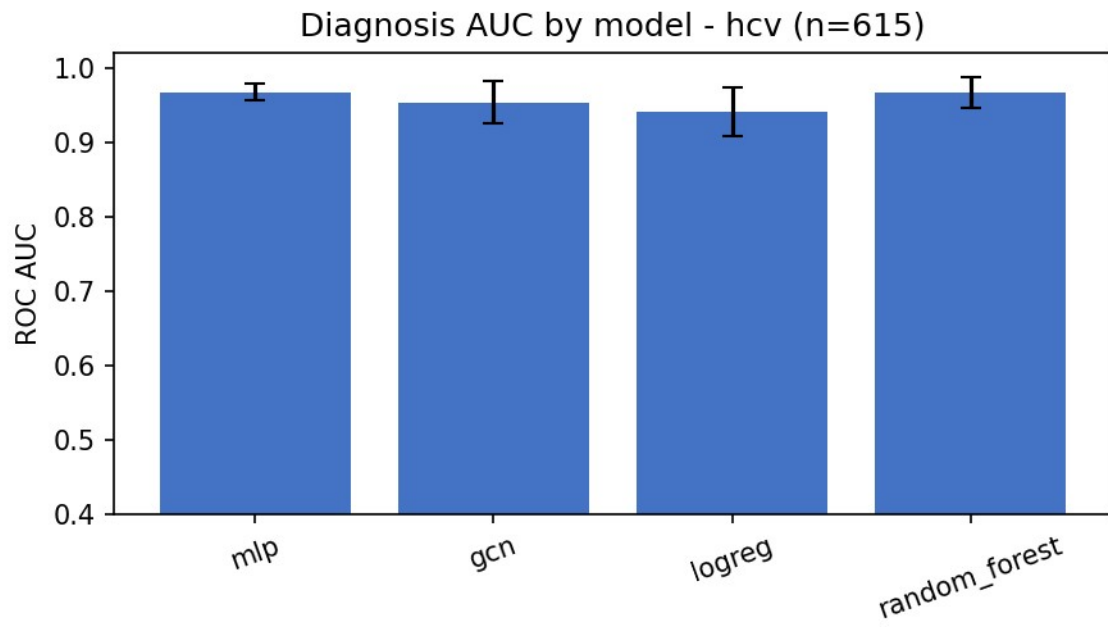
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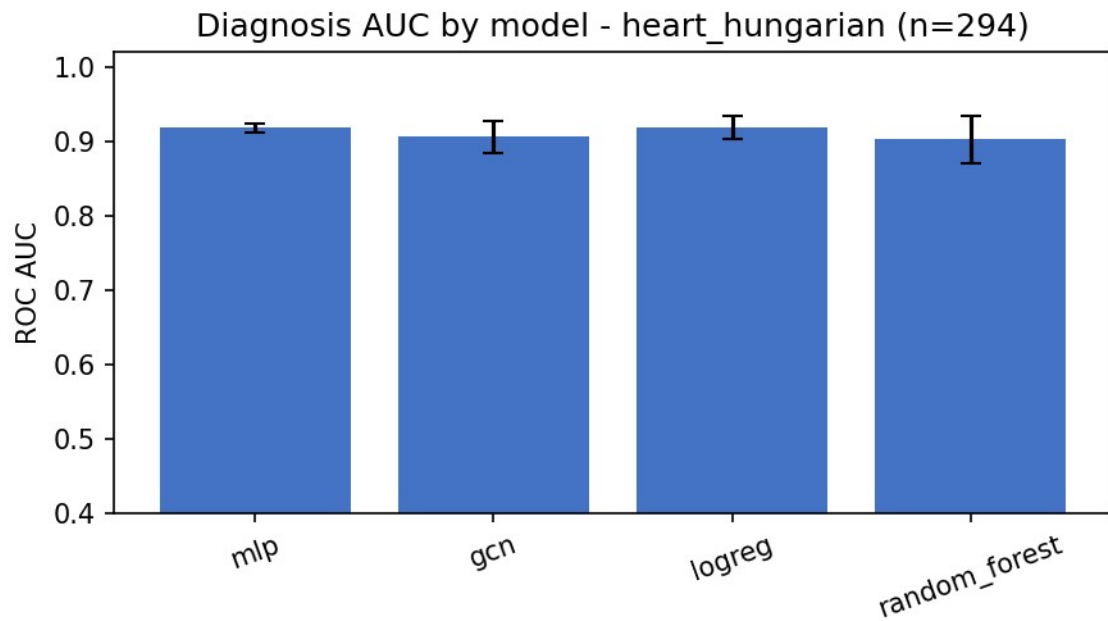
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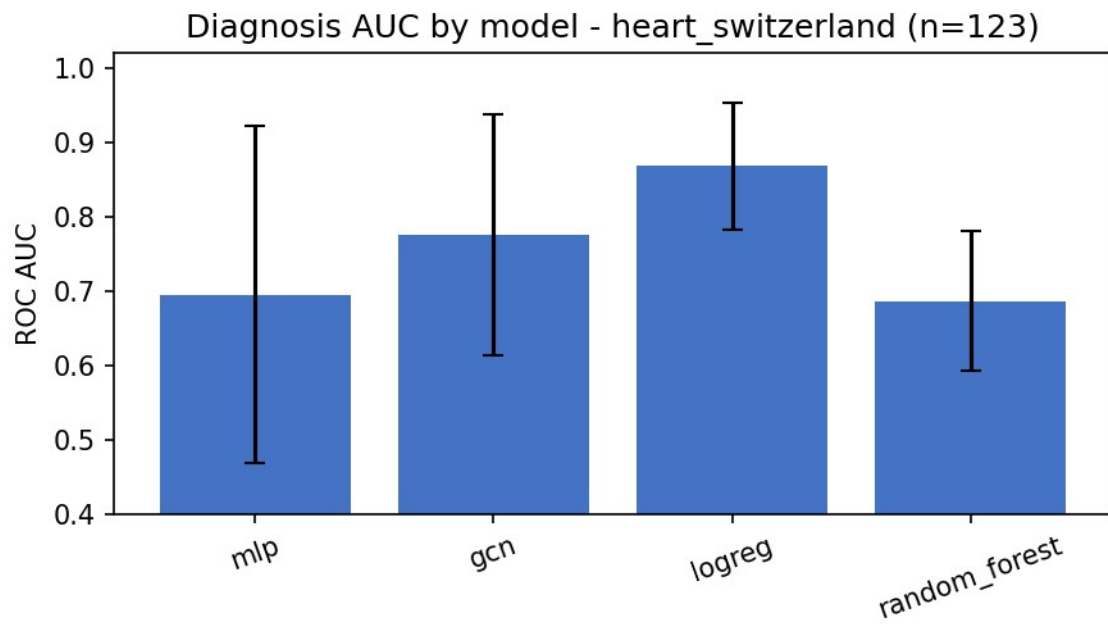
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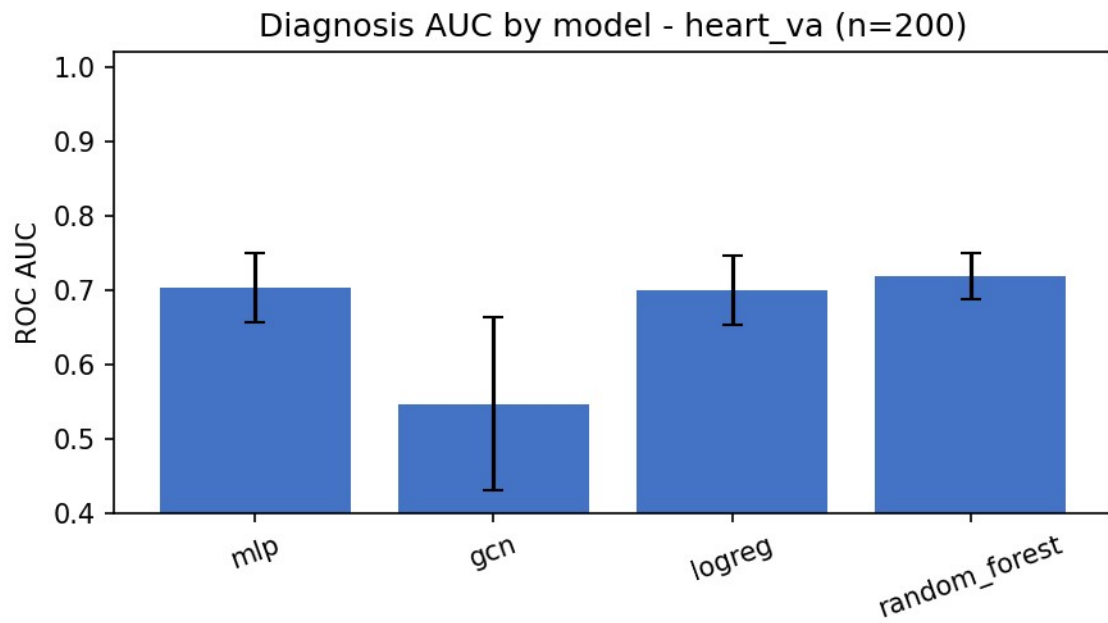
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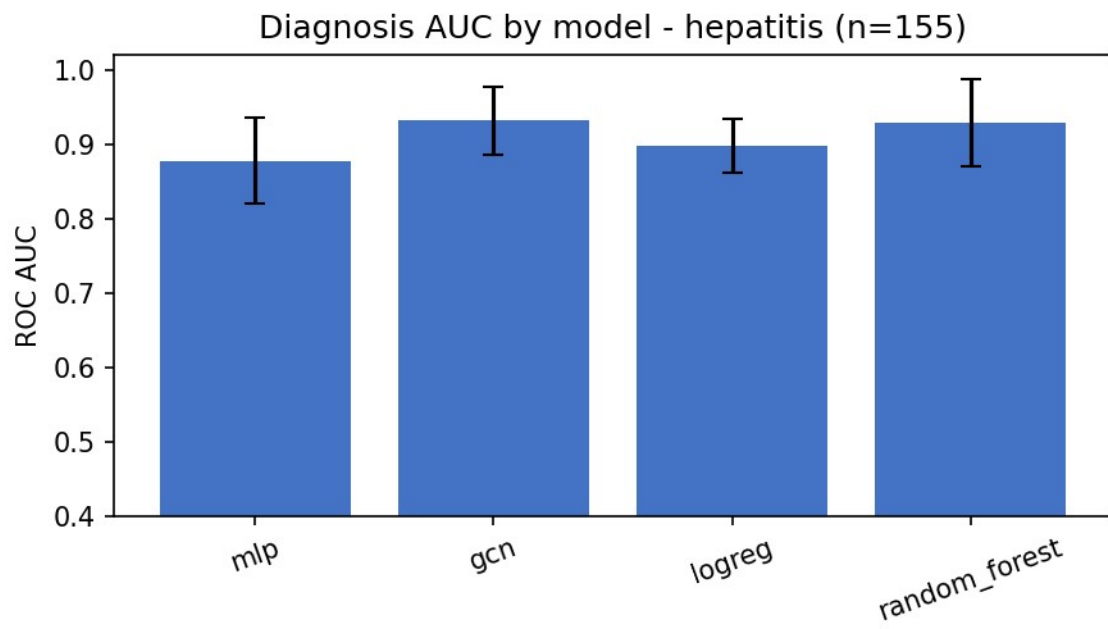
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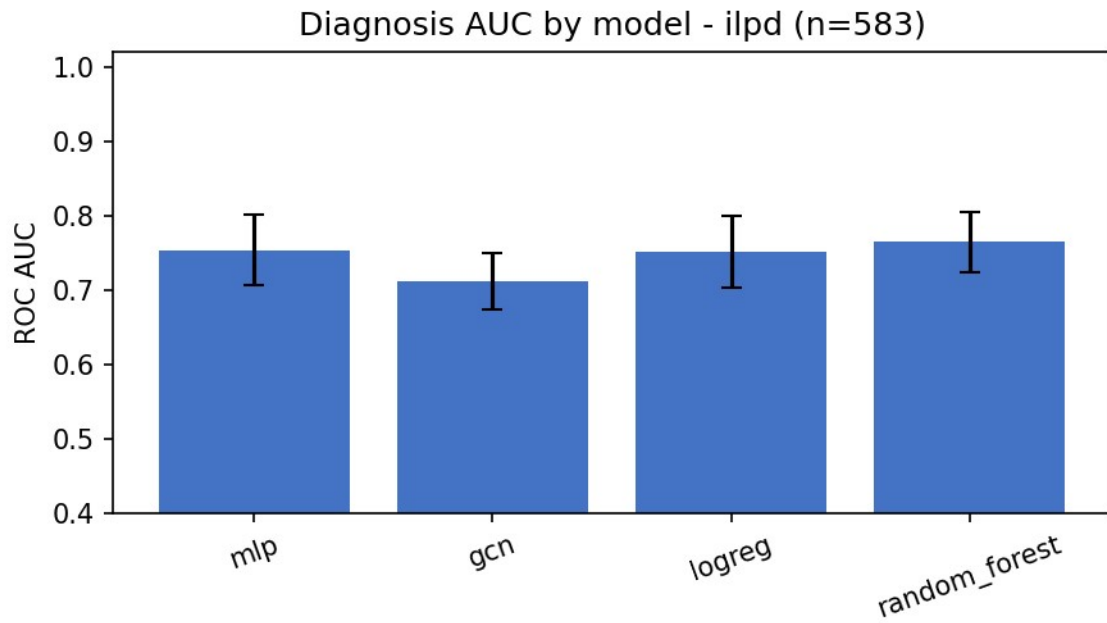
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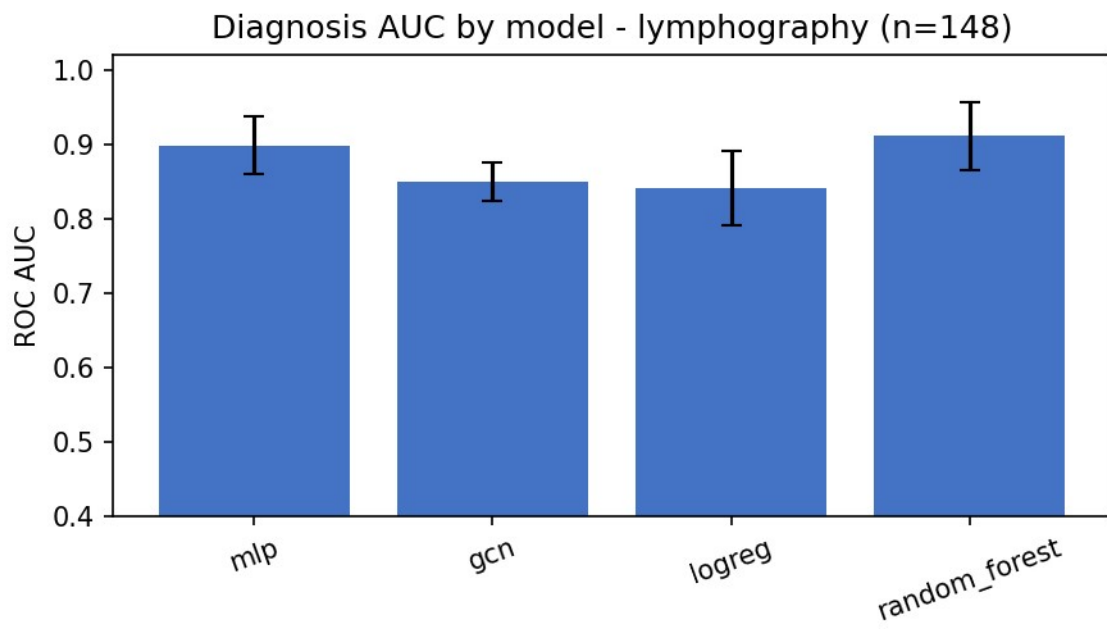
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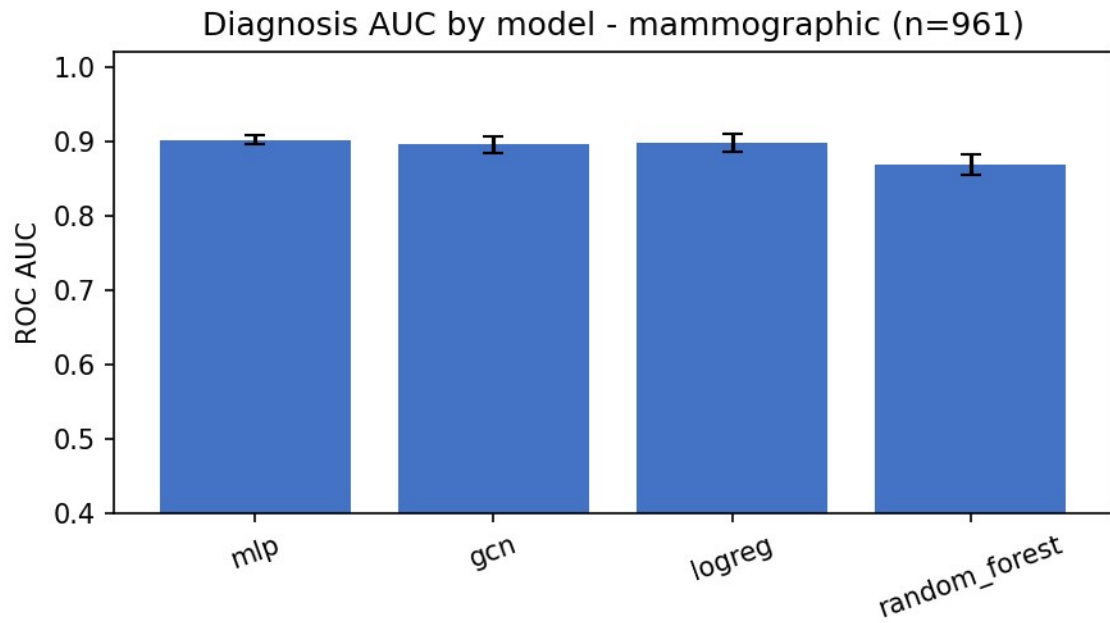
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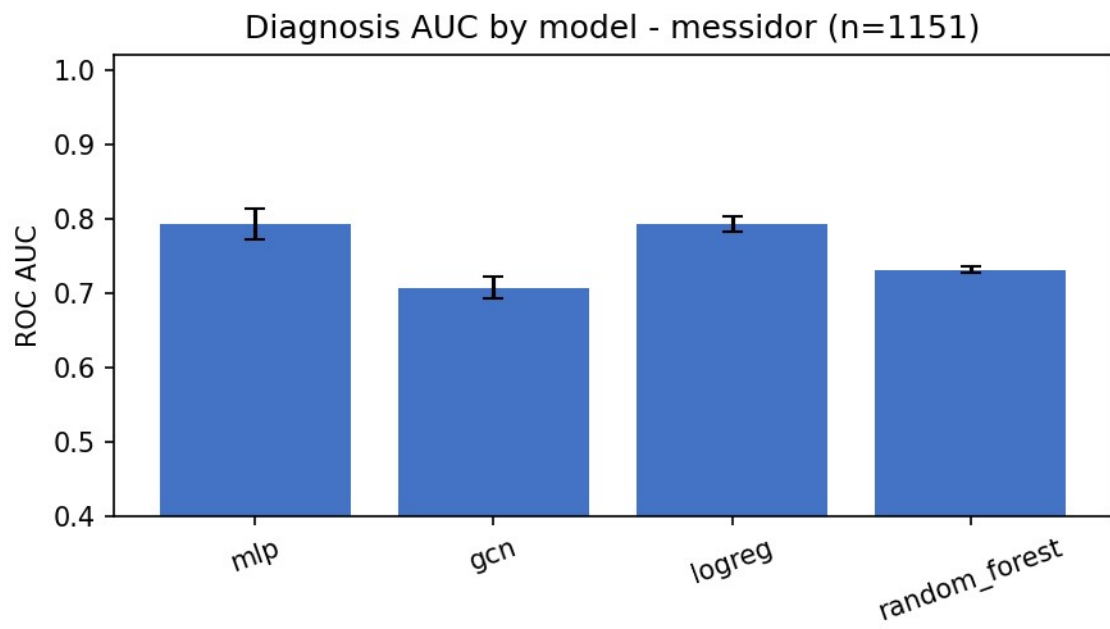
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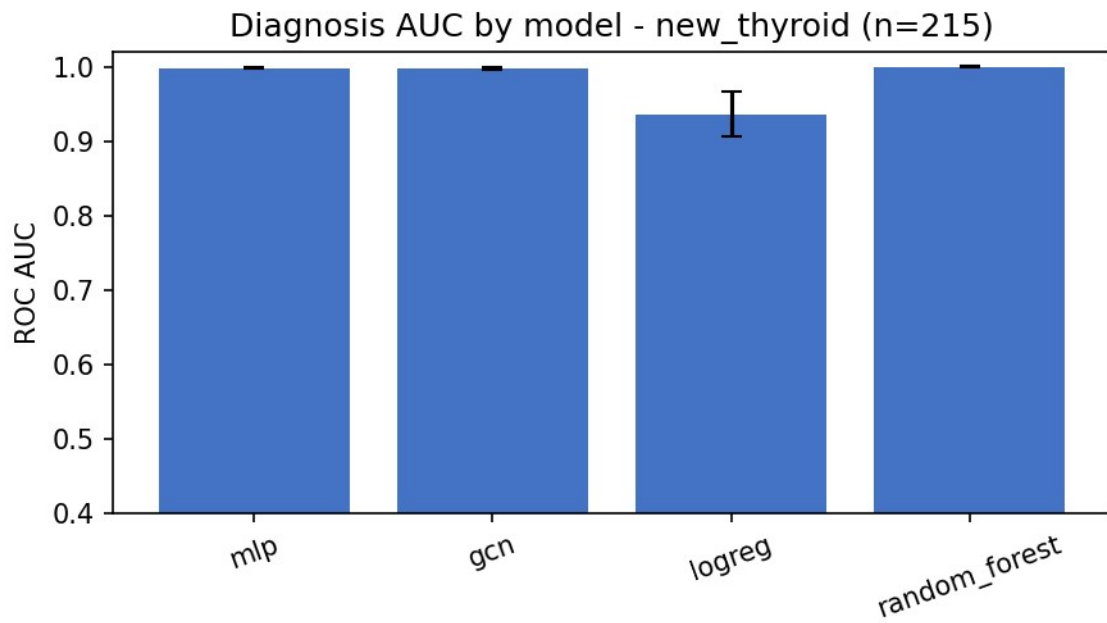
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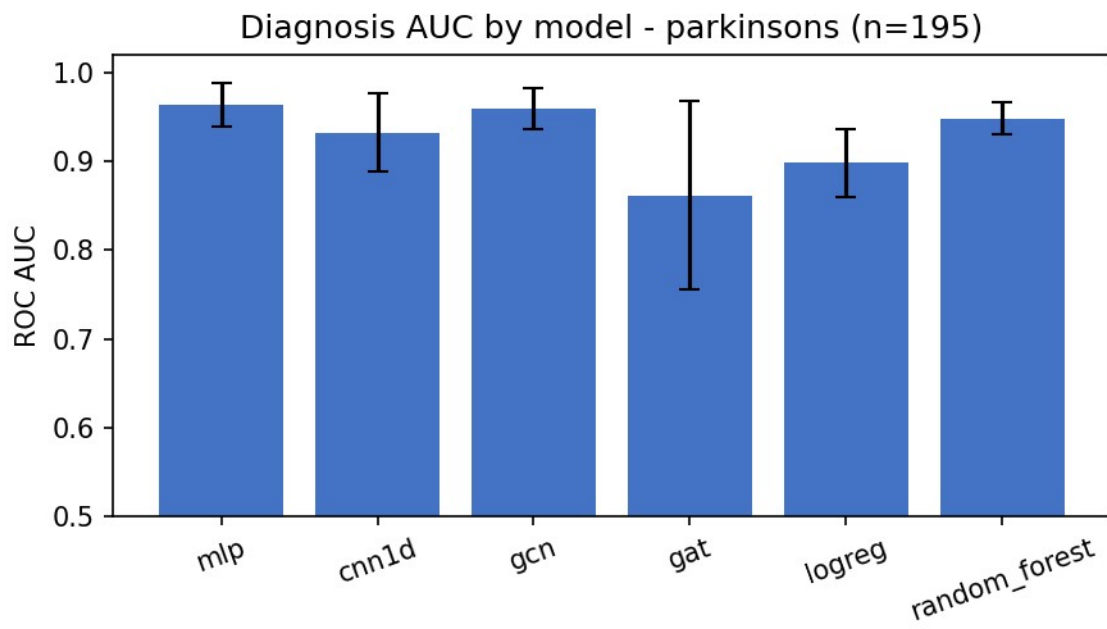
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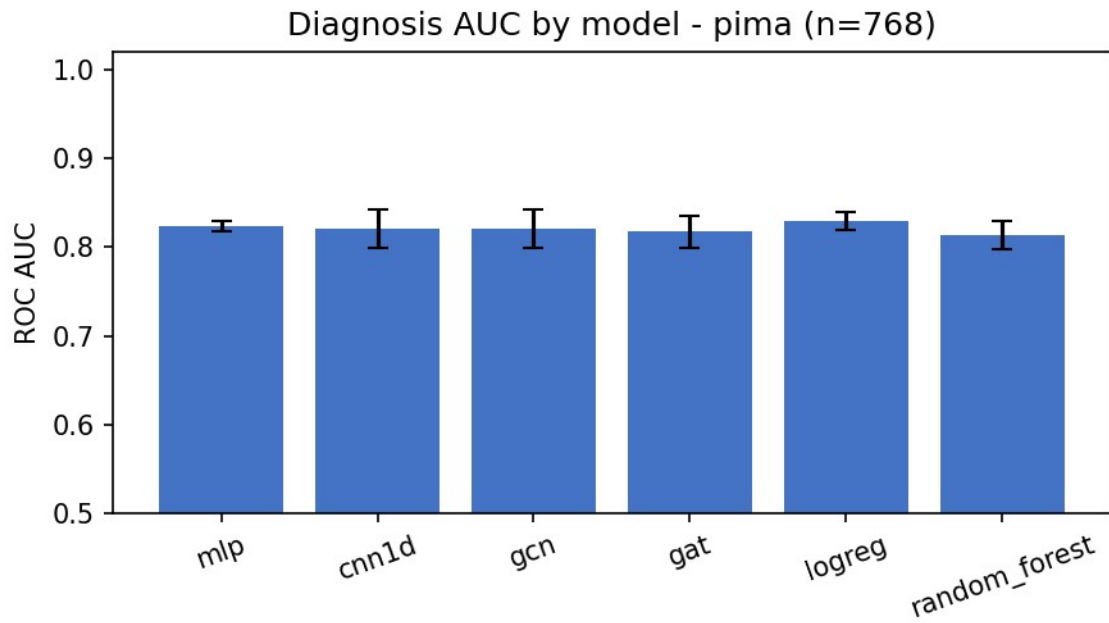
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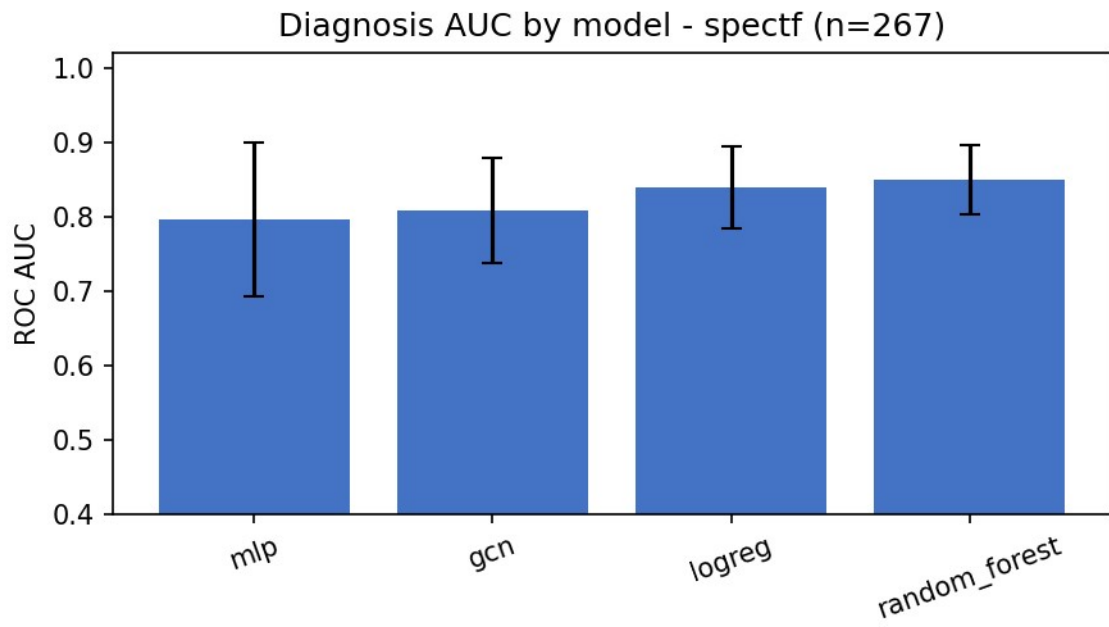
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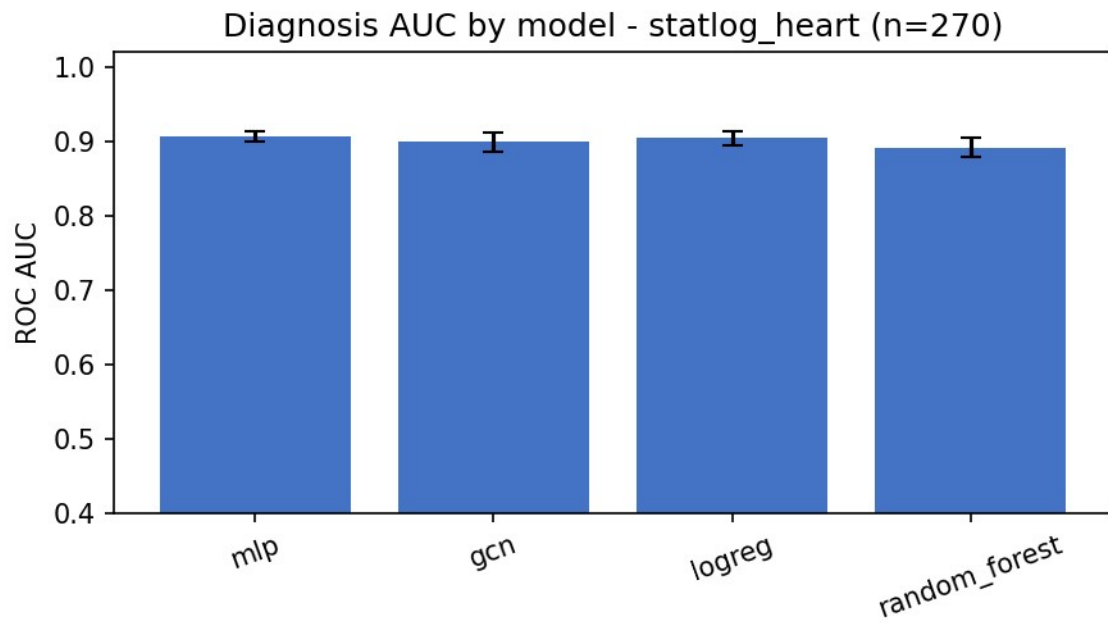
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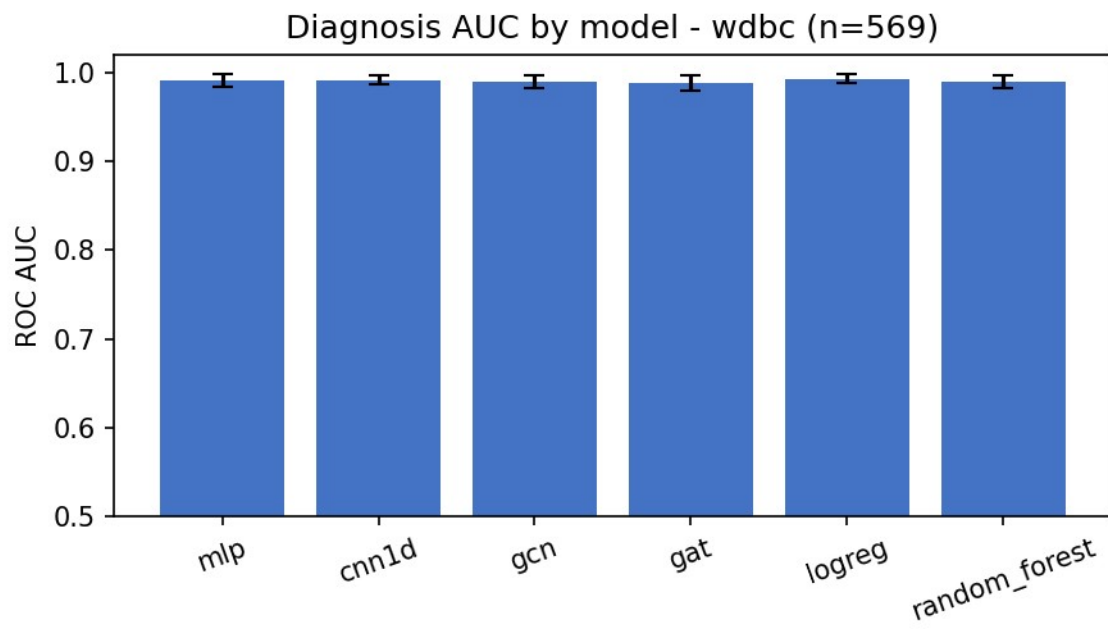
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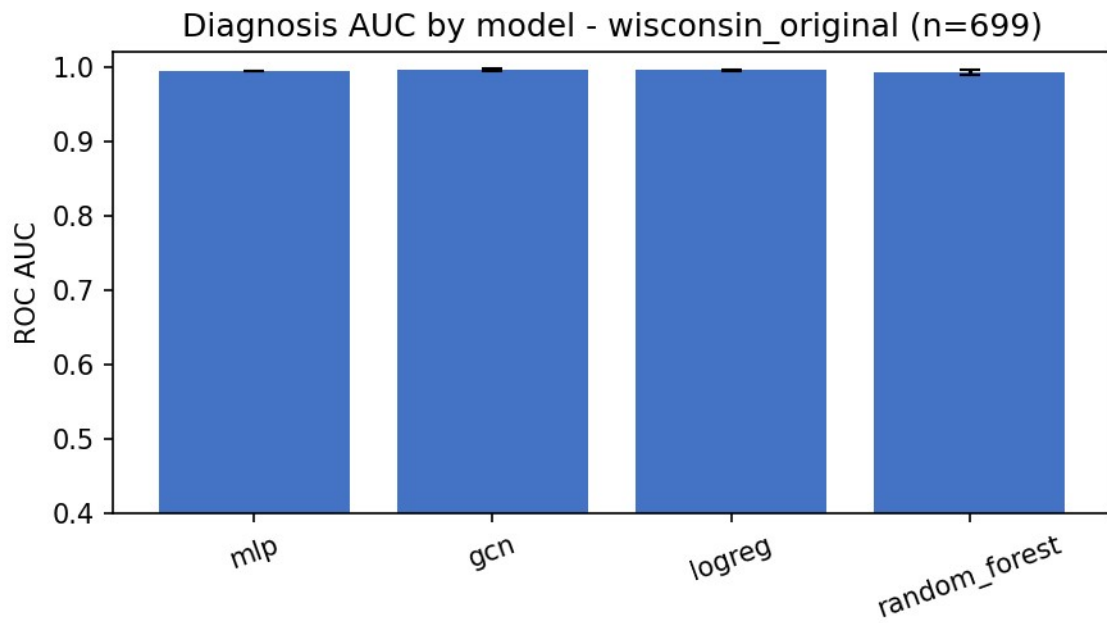
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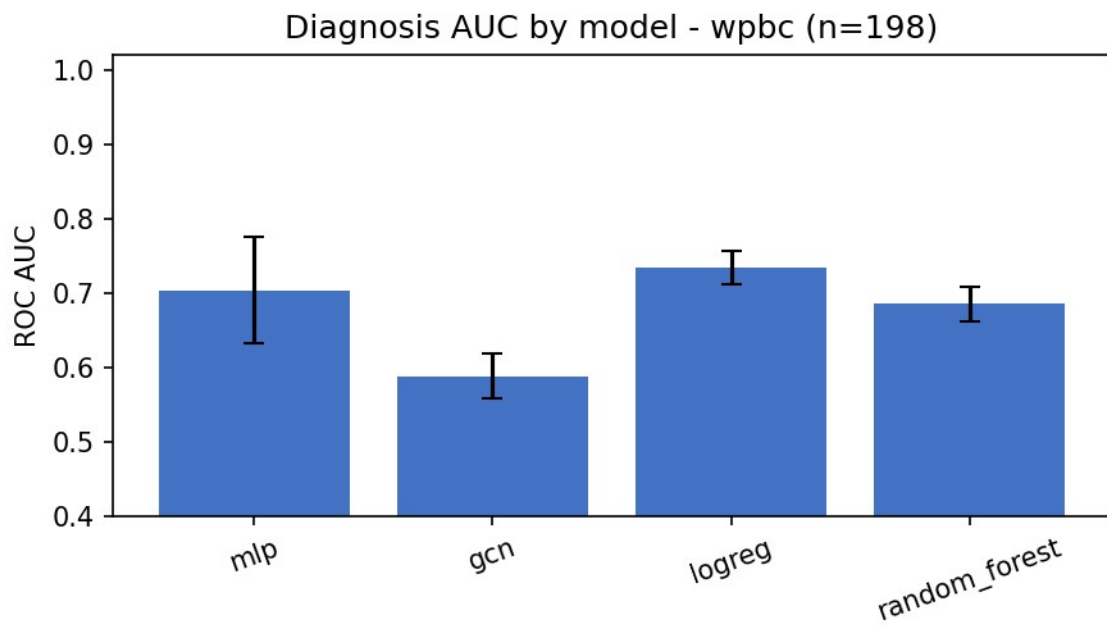
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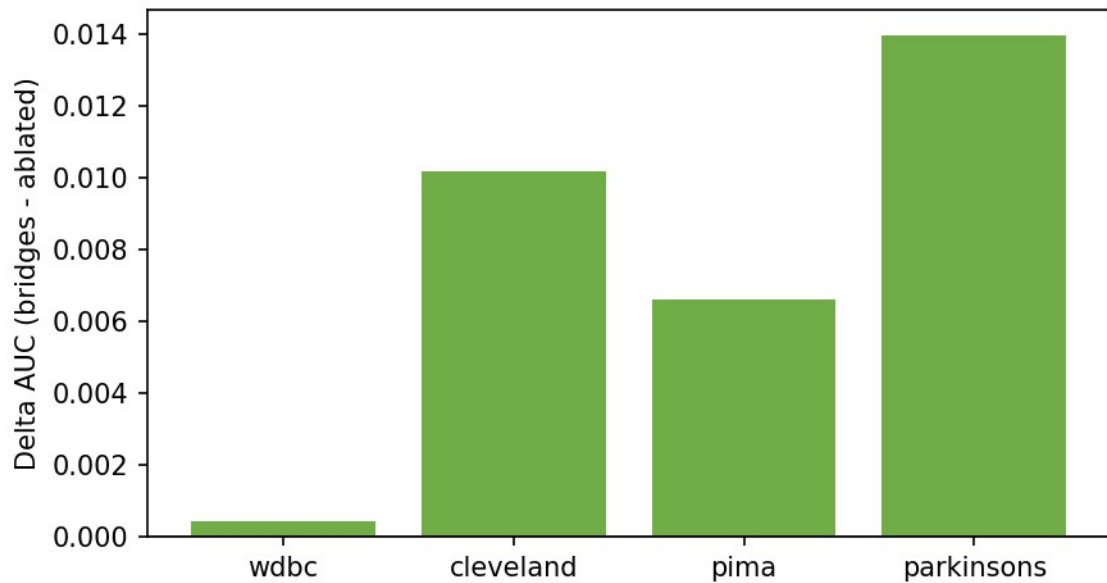


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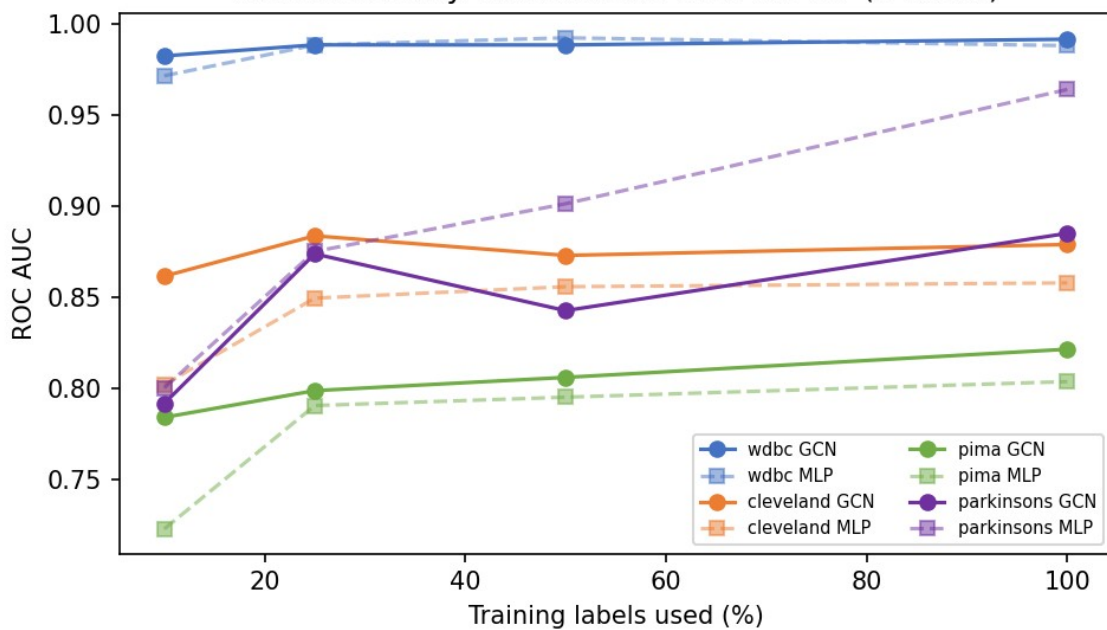
auc wpbc

Cross-disease bridge effect on diagnosis AUC (mean over 5 seeds)



bridge delta

Label efficiency: transductive GCN vs MLP (5 seeds)



label efficiency